

Long bones like this one are also found in your arms.

These blood vessels supply oxygen and nutrients to the bone and remove waste.

A tough membrane protects the bone's surface and provides an anchor for your muscles.

The shaft links the bone's two rounded heads.

Bone tissue contains calcium, a substance obtained from food such as milk and cheese.

WOW!

Your red bone marrow makes red blood cells at a rate of about 2 million cells every second.

BUILT FOR STRENGTH

Your bones are far stronger and lighter than concrete or steel. If your skeleton was made from steel it would weigh five times as much, making it almost impossible for you to move around. The complex honeycomblike design of tiny struts and hollow spaces inside bones has been copied in many man-made structures.

The head of the thighbone carries the body's weight.

These criss-cross struts in spongy bone are arranged in a way that provides maximum support.

Compact bone is tough and dense so that it doesn't buckle under stress.

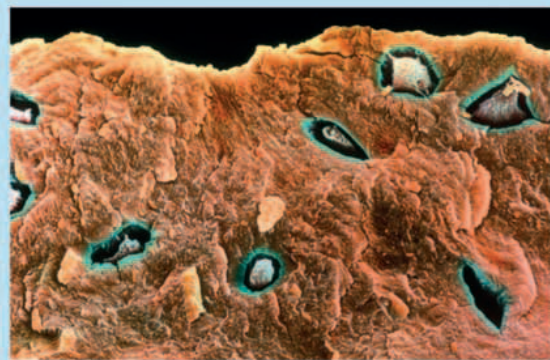
◀ **THIGHBONE** The arched shape of the thighbone, and the arrangement of struts inside it, enable it to help support the body.

▲ **EIFFEL TOWER, FRANCE** The arch at the base of this tower supports its weight, helped by the criss-cross pattern of metal struts.

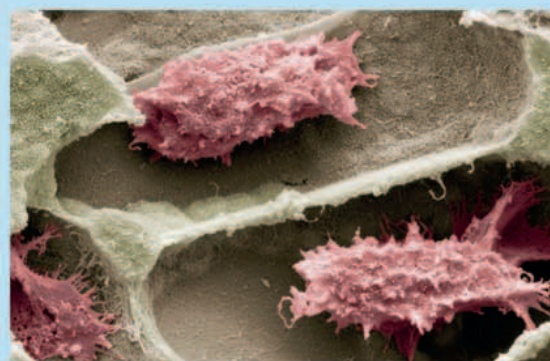


LOOK CLOSER: BUILD OR BREAK

Your bones are constantly being broken down and built up to make sure they are as strong as possible. Two types of cells are responsible. These cells are known as osteoblasts and osteoclasts.



▲ **OSTEOBLASTS** Found on the surface of the bone, osteoblasts are bone-building cells. They are surrounded here by the bone tissue (orange) they have just made.



▲ **OSTEOCLASTS** These cells (pink) break down worn-out bone. Here they are busy eating away at the surrounding tissue (cream).

